Simple Types

Concepts of Programming Languages

CAS CS 320

Outline

- Have a high-level discussion of **type theory** in general
- Introduce and analyze the **simply-typed lambda calculus** (STLC)
- Demo an **implementation** of the STLC

Recap

Recall: The Environment Model

$$\mathscr{E} \vdash e \Downarrow v \quad \textit{or} \quad \langle \mathscr{E}, e \rangle \Downarrow v$$

Idea. We keep track of their values in an *environment*

We evaluate *relative* to the environment and *lazily* fill in variable values along the way

Recall: Closures

$$(\mathscr{E}, e)$$

Def. *(informal)* A **closure** is an expression plus an environment

It *captures* bindings it needs to evaluate its expression

Idea. Functions need to *remember* what the environment looked like in order to behavior correctly (according to lexical scoping)

Recall: Named Closures

$$(\mathscr{E}, \mathsf{name} \mapsto \lambda x\,.\,e)$$

We need to be able to *name* closures

Idea. Named closures will put themselves into their environment *when they're called*

Recall: Lambda Calculus$^+$ (Syntax)

```
<expr> ::= λ<var>.<expr>
         | <var>
         | <expr><expr>
         | let <var> = <expr>
           in <expr>
         | <num>

<val>  ::= λ<var>.<expr>
         | <num>
```

Example (Dynamic)

$$\frac{}{\mathscr{E} \vdash \lambda x\,.\,e \Downarrow \lambda x\,.\,e} \qquad \frac{}{\mathscr{E} \vdash n \Downarrow n} \qquad \frac{\mathscr{E}(x) = v}{\mathscr{E} \vdash x \Downarrow v}$$

$$\frac{\mathscr{E}_1 \vdash e_1 \Downarrow v_1 \qquad \mathscr{E}_2 = \mathscr{E}_1[x \mapsto v_1] \qquad \mathscr{E}_2 \vdash e_2 \Downarrow v_2}{\mathscr{E}_1 \vdash \mathsf{let}\ x = e_1\ \mathsf{in}\ e_2 \Downarrow v_2}$$

$$\frac{\mathscr{E} \vdash e_1 \Downarrow \lambda x\,.\,e \qquad \mathscr{E} \vdash e_2 \Downarrow v_2 \qquad \mathscr{E}' = \mathscr{E}[x \mapsto v_2] \qquad \mathscr{E}' \vdash e \Downarrow v}{\mathscr{E} \vdash e_1 e_2 \Downarrow v}$$

$$\emptyset \ \vdash\ \mathsf{let}\ x = 0\ \mathsf{in}\ \mathsf{let}\ f = \lambda y\,.\,x\ \mathsf{in}\ \mathsf{let}\ x = 1\ \mathsf{in}\ f\ 0\ \Downarrow\ 1$$

Example (Lexical)

$$\frac{}{\mathscr{E} \vdash \lambda x\,.\,e \Downarrow (\mathscr{E}, \lambda x\,.\,e)} \qquad \frac{}{\mathscr{E} \vdash n \Downarrow n} \qquad \frac{\mathscr{E}(x) = v}{\mathscr{E} \vdash x \Downarrow v}$$

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$$\emptyset \;\vdash\; \text{let } x = 0 \text{ in let } f = \lambda y\,.\,x \text{ in let } x = 1 \text{ in } f\ 0 \;\Downarrow\; 1$$

Practice Problem

```
let x = 0 in
let g = fun y -> x + 1 in
let x = 1 in
let f = fun y -> g x in
let x = 2 in
f
```

What (closure) does the following expression evaluate to? You don't need to give the derivation

Answer

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let x = 0 in
let g = fun y -> x + 1 in
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```

Type Theory

What is a Type?

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Types help us delineate "well-behaved" programs

Trade-offs

$$(\lambda x\,.\,xx)(\lambda x\,.\,xx)$$

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- Simplicity/Usability
- Expressivity
- Safety/Theoretical Guarantees

OCaml

```
# let big_omega =
    let little_omega x = x x in
    little_omega little_omega;;
Error: This expression has type 'a -> 'b
       but an expression was expected of type 'a
       The type variable 'a occurs inside 'a -> 'b
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More expressive means more complex, designing PLs is finding the balance

Simply-Typed Lambda Calculus

Recall: Typing Judgments

$$\Gamma \vdash e : \tau$$

This judgment reads:

e has type τ in the context Γ

We say that e is **well-typed** if $\cdot \vdash e : \tau$ for some type τ

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We say that e is **well-typed** if $\cdot \vdash e : \tau$ for some type τ

Most of what type theorists do is come up with rules for deriving typing judgments

STLC Syntax

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<e>  ::= () | <v> | <e> <e>
       | fun ( <v> : <ty> ) -> <e>
<ty> ::= unit | <ty> -> <ty>
<v>  ::= a | ... | z
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Syntax

$$e ::= \bullet \mid x \mid \lambda x^{\tau}.\, e \mid ee$$

$$\tau ::= \top \mid \tau \to \tau$$

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These rules enforce that a function can only be applied if we *know* that it's a function

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If we include annotations we're using **Church-style typing.** If we drop annotations, we're using **Curry-style typing**

Aside: Church vs. Curry Typing

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fun x -> x

fun (x : unit) -> x
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(using Curry-style typing is not the same as having polymorphism)

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Every expression has a *unique type* in STLC with Church-style typing

In particular, the function `type_of` is well-defined

STLC Semantics (Review)

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This is part of the point: Type-checking only determines *whether* we evaluate the program (whether it makes sense to)

It doesn't determine **how** we evaluate the program

Example

$$\lambda x^{\tau}.xx$$

What happens if we try to give a type to the above expression? What should τ be?

Practice Problem

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau} \qquad \frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}$$

$\cdot \vdash \lambda f^{\top \to \top}.\, \lambda x^{\top}.\, fx : (\top \to \top) \to \top \to \top$

Give a derivation for the above judgment

Answer

$$\cdot \vdash \lambda f^{\top \to \top}.\lambda x^{\top}.fx : (\top \to \top) \to \top \to \top$$

How do we know if we've defined a "good" programming language?

Type Safety

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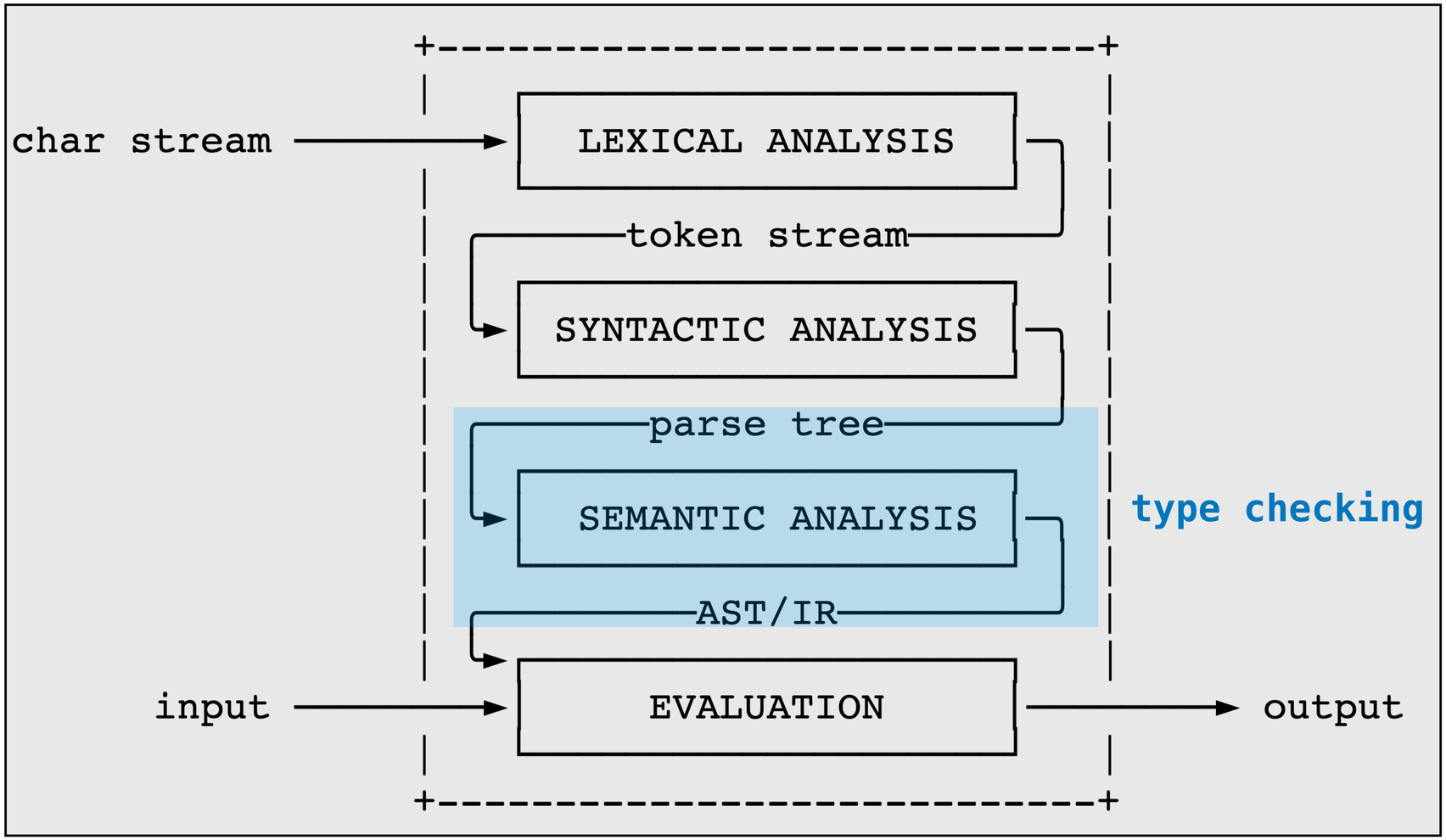
These results are *fundamental*. They tell us that our PL is well-behaved (it's a "good" PL)

Type Checking

The Picture


char stream
LEXICAL ANALYSIS
token stream
SYNTACTIC ANALYSIS
parse tree
SEMANTIC ANALYSIS
type checking
AST/IR
input
EVALUATION
output

Type Checking vs. Type Inference

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For STLC, they are both easy

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Our solution: We'll just use type inference

demo

(STLC)

Summary

Type systems delineate well-behaved expressions